

二次域 练习题

设 $D \in \mathbb{Z}$ 是无平方因子的非零整数, $D \neq 1$

(ii) 说明 $\mathbb{Q}(\sqrt{D})$ 的元素都能唯一地表示为 $a+b\sqrt{D}$ 的形式.

$$\text{Proof: } \mathbb{Q}(\sqrt{D}) = \{a+b\sqrt{D} \in \mathbb{C} : a, b \in \mathbb{Q}\}$$

对 $\forall a_1+b_1\sqrt{D}, a_2+b_2\sqrt{D} \in \mathbb{Q}(\sqrt{D})$ (其中 $a_1, b_1, a_2, b_2 \in \mathbb{Q}$)

若 $a_1+b_1\sqrt{D} = a_2+b_2\sqrt{D}$, 则有: $a_1-a_2 = (b_2-b_1)\sqrt{D}$.

假设 $b_2-b_1 \neq 0$, 则有: $\sqrt{D} = \frac{a_1-a_2}{b_2-b_1}$

$$\therefore a_1-a_2 \in \mathbb{Q}, b_2-b_1 \in \mathbb{Q}, b_2-b_1 \neq 0 \quad \therefore \sqrt{D} = \frac{a_1-a_2}{b_2-b_1} \in \mathbb{Q}$$

若 $D > 0$, 则有: $D \in \mathbb{Z}$ 是无平方因子的非零正整数, $D \neq 1$

$\therefore \sqrt{D} \in \mathbb{R} \setminus \mathbb{Q}$. 矛盾.

若 $D < 0$, 则有: $\sqrt{D} = \sqrt{-(-D)} = \sqrt{(-1)(-D)} = \sqrt{-1} \cdot \sqrt{-D} = \sqrt{-D} \cdot i \notin \mathbb{Q}$.

矛盾.

$$\therefore b_2-b_1 = 0. \quad \therefore b_1 = b_2$$

$$\therefore a_1-a_2 = (b_2-b_1)\sqrt{D} = 0 \cdot \sqrt{D} = 0 \quad \therefore a_1 = a_2$$

$\therefore a_1 = a_2$ 且 $b_1 = b_2$ 唯一性得证. $\square$

(iii) 验证 $\mathbb{Z}[\sqrt{D}] := \{a+b\sqrt{D} : a, b \in \mathbb{Z}\}$ 是整环.

Proof: $\because \mathbb{Q}(\sqrt{D})$ 是 $\mathbb{C}$ 的子域 $\therefore \mathbb{Q}(\sqrt{D})$ 是域 $\therefore \mathbb{Q}(\sqrt{D})$ 是整环

$\mathbb{Q}(\sqrt{D})$ 是环, $\mathbb{Z}[\sqrt{D}] \subseteq \mathbb{Q}(\sqrt{D})$

$$\therefore 0 = 0 + 0\sqrt{D} \in \mathbb{Z}[\sqrt{D}] \quad \therefore \mathbb{Z}[\sqrt{D}] \neq \emptyset$$

$$0 = 0 + 0\sqrt{D} \in \mathbb{Z}[\sqrt{D}], \quad 1 = 1 + 0\sqrt{D} \in \mathbb{Z}[\sqrt{D}]$$

对 $\forall a+b\sqrt{D}, c+d\sqrt{D} \in \mathbb{Z}[\sqrt{D}]$ (其中 $a, b, c, d \in \mathbb{Z}$), 有:

$$(a+b\sqrt{D}) + (c+d\sqrt{D}) = (a+c) + (b+d)\sqrt{D} \in \mathbb{Z}[\sqrt{D}]$$

$$(a+b\sqrt{D})(c+d\sqrt{D}) = (ac+bdD) + (ad+bc)\sqrt{D} \in \mathbb{Z}[\sqrt{D}]$$

$$\because a+b\sqrt{D} \in \mathbb{Z}[\sqrt{D}] \quad \therefore (-a) + (-b)\sqrt{D} \in \mathbb{Z}[\sqrt{D}]$$

$$\therefore (a+b\sqrt{D}) + ((-a) + (-b)\sqrt{D}) = 0 + 0\sqrt{D} = 0$$

$$\therefore -(a+b\sqrt{D}) = (-a) + (-b)\sqrt{D} \in \mathbb{Z}[\sqrt{D}]$$

$\therefore \mathbb{Z}[\sqrt{D}]$ 是 $\mathbb{Q}(\sqrt{D})$ 的子环. $\therefore \mathbb{Z}[\sqrt{D}]$ 是环, 且是非零环

$\therefore$ 复数乘法满足交换律 $\therefore \mathbb{Z}[\sqrt{D}]$ 是交换环

对 $\forall a+b\sqrt{D}, c+d\sqrt{D} \in \mathbb{Z}[\sqrt{D}]$ (其中 $a, b, c, d \in \mathbb{Z}$),

若 $(a+b\sqrt{D}) \cdot (c+d\sqrt{D}) = 0$, 则有: $a+b\sqrt{D} \in \mathbb{C}, c+d\sqrt{D} \in \mathbb{C}$, 且

$$(a+b\sqrt{D})(c+d\sqrt{D}) = 0. \quad \therefore \mathbb{C} \text{ 是域} \quad \therefore \mathbb{C} \text{ 是整环}$$

$$\therefore a+b\sqrt{D} = 0 \text{ 或 } c+d\sqrt{D} = 0 \quad \therefore \mathbb{Z}[\sqrt{D}] \text{ 是整环.}$$

(或这样证: $\because \mathbb{Q}(\sqrt{D})$ 是域 $\therefore \mathbb{Q}(\sqrt{D})$ 是整环.

$\therefore \mathbb{Z}[\sqrt{D}]$ 是 $\mathbb{Q}(\sqrt{D})$ 的子环 $\therefore \mathbb{Z}[\sqrt{D}]$ 是整环.) $\square$

(iv) $O_D = \left\{ \frac{x}{2} + \frac{y}{2}\sqrt{D} \mid x, y \in \mathbb{Z} \text{ 且 } x \equiv y \pmod{2} \right\}$ 是整环, 当 $D \equiv 1 \pmod{4}$ 时.

Proof: $\mathbb{Q}(\sqrt{D})$ 是环, $O_D \subseteq \mathbb{Q}(\sqrt{D})$

$$0 = \frac{0}{2} + \frac{0}{2}\sqrt{D} \in O_D, \quad 1 = \frac{2}{2} + \frac{0}{2}\sqrt{D} \in O_D \quad \therefore O_D \neq \emptyset.$$

对 $\forall \frac{x}{2} + \frac{y}{2}\sqrt{D}, \frac{\alpha}{2} + \frac{\beta}{2}\sqrt{D} \in O_D$ (其中 $x, y, \alpha, \beta \in \mathbb{Z}, x \equiv y \pmod{2}, \alpha \equiv \beta \pmod{2}$)

$\alpha \equiv \beta \pmod{2}$)